Example 3.16 If $\sin x = \frac{4}{5}$ (in I quadrant) and $\cos y = \frac{-12}{13}$ (in II quadrant), then find (i) $\sin(x-y)$, (ii) $\cos(x-y)$.

Solution:
Given that $\sin x = \frac{4}{5}$.

$$\cos^2 x + \sin^2 x = 1 \text{ gives } \cos x = \pm\sqrt{1-\sin^2 x} = \pm\sqrt{1-\frac{16}{25}} = \pm\frac{3}{5}$$

In the first quadrant, $\cos x$ is always positive. Thus, $\cos x = \frac{3}{5}$.
Also, given that $\cos y = -\frac{12}{13}$ in the II quadrant. We have

$$\sin y = \pm\sqrt{1-\cos^2 y} = \pm\sqrt{1-\frac{144}{169}} = \pm\frac{5}{13}.$$

In the second quadrant, $\sin y$ is always positive. Thus, $\sin y = \frac{5}{13}$.

(i) $\sin(x-y) = \sin x\cos y - \cos x \sin y = \frac{4}{5}\left(\frac{-12}{13}\right) - \frac{3}{5}\left(\frac{5}{13}\right) = -\frac{63}{65}$.

(ii) $\cos(x-y) = \cos x\cos y + \sin x \sin y = \frac{3}{5}\left(\frac{-12}{13}\right) + \frac{4}{5}\left(\frac{5}{13}\right) = -\frac{16}{65}$.

Example 3.17 Prove that $\cos\left(\frac{3\pi}{4}+x\right) - \cos\left(\frac{3\pi}{4}-x\right) = -\sqrt{2}\sin x$

Solution:

$$\begin{aligned}\text{L.H.S} &= \cos\frac{3\pi}{4}\cos x - \sin\frac{3\pi}{4}\sin x - \cos\frac{3\pi}{4}\cos x - \sin\frac{3\pi}{4}\sin x \\ &= -2\sin\left(\pi-\frac{\pi}{4}\right)\sin x = -2\left(\frac{1}{\sqrt{2}}\right)\sin x = -\sqrt{2}\sin x.\end{aligned}$$

Observe that $\cos(A+x) - \cos(A-x) = -2\sin A \sin x$

Example 3.18 Point $A(9,12)$ rotates around the origin O in a plane through 60° in the anticlockwise direction to a new position B. Find the coordinates of the point B.

Solution:
Let $A(9,12) = A(r\cos\theta, r\sin\theta)$, where $r = OA$. Then $r\cos\theta = 9$ and $r\sin\theta = 12$.
Thus, $r^2 = 81 + 144 = 225 \Rightarrow r = 15$.
Hence, the point A is given by $A\,(15\cos\theta, 15\sin\theta)$.
Now, the point B is given by $B\,(15\cos(\theta+60^\circ), 15\sin(\theta+60^\circ))$.

$$\begin{aligned}15\cos(\theta+60^\circ) &= 15\,(\cos\theta\cos 60^\circ - \sin\theta\sin 60^\circ) \\ &= (15\cos\theta)\cos 60^\circ - (15\sin\theta)\sin 60^\circ = 9\times\frac{1}{2} - 12\times\frac{\sqrt{3}}{2} = \frac{3}{2}(3-4\sqrt{3})\end{aligned}$$